PHYS7101: NUMERICAL METHODS AND DATA ANALYSIS · LECTURE 8

Updating what we know

Preparing and relating measured signals;
Bayesian inference and Markov chain Monte Carlo

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Where we left off

From earlier lectures

- Interpolation (L4); MAD, bootstrap, surrogates (L5)
- Maximum likelihood; Poisson counts of radioactive decay, $\tau = 20\,\mathrm{s}$ (L3)
- A distribution family only when physics supplies it (L5)
- Monte Carlo (L2, L5); autocorrelation (L6); correlation (L7)

Today

A Two measured signals: preparing and relating them

B Bayes' theorem: likelihood, prior, posterior; credible intervals

C Sampling the posterior: Metropolis–Hastings

D Outlook: parameters that change in time

Part A analyses the data without assuming a model of how they arise; Parts B–D assume one. Bayesian inference needs such a model: use it when physics supplies it, as for a parametric description in Lecture 5.

PART A

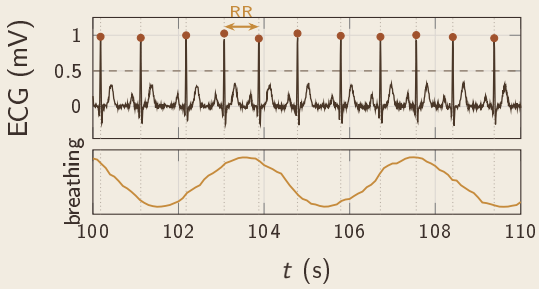
Two measured signals

Heartbeat and breathing: preparing the data, relating the signals, and the size of a change, with methods that assume no model of how the data arise.

Heart and breathing: what do we record?

Problem: the heart and the lungs may interact: breathing in speeds the heart up, breathing out slows it down. To study their interaction, we first record the activity of each. What do we record, and how do we get the heartbeats?

ECG and breathing, recorded simultaneously



- **ECG**: its **R peaks** above a threshold (dashed) give the beat times t_j.
- **Breathing**: a continuous signal (chest belt).

How the signals are generated

$$\dot{\varphi}_r = 2\pi \times 0.25\,\text{Hz} + \varepsilon_r(t)$$

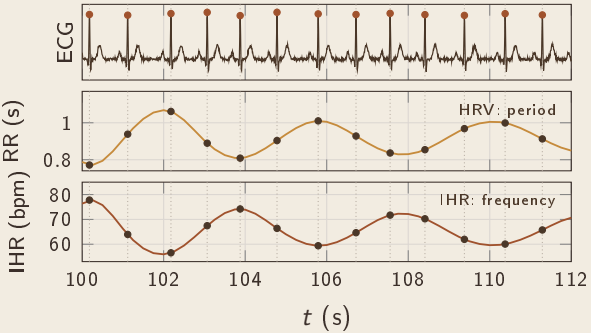
$$\dot{\varphi}_h = 2\pi \times 1.1\,\text{Hz} + K\cos\varphi_r + \varepsilon_h(t)$$

φ_r, φ_h: phases of respiration and heart, growing by 2π per cycle (0.25 Hz: a breath every 4 s; 1.1 Hz: ≈ 66 beats/min); ε: noise. Breathing: $\cos\varphi_r$, rising as we breathe in; a beat each time φ_h completes a cycle. $K = 0.8$ rad/s: breathing in speeds the heart up, out slows it ($\approx \pm 12\%$); one direction only.

Data: simulated by the model on the right (not used by the analysis); fixed ECG beat shape.

From heartbeats to heart rate in time

Problem: each heartbeat is a cycle; the R peak marks one moment in it. Knowing only these moments, how do we follow the heart rate in time?



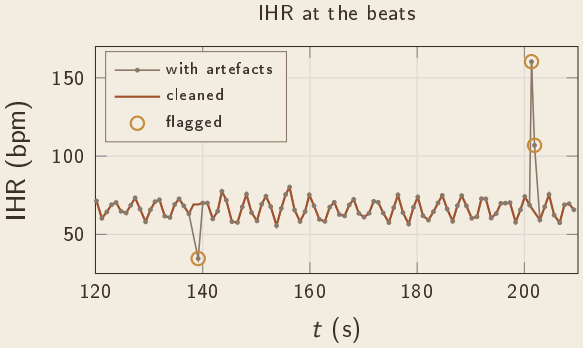
1. beat times t_j from the R peaks
2. $\mathrm{RR}_j = t_j - t_{j-1}$: the period of each cycle; in time, the **HRV**
3. $\mathrm{IHR}_j = 60/\mathrm{RR}_j$ (beats/min, bpm): its frequency
4. both known once per cycle, at uneven t_j (dots): interpolate (L4) to an even 4-Hz grid, well above the beat rate (≈ 1.1 Hz); cubic spline (line)

Caution: the IHR rises and falls every ≈ 4 s: breathing modulates the heart rate (**respiratory sinus arrhythmia**). With ≈ 4.4 values per rise and fall, linear interpolation flattens it (L4): standard deviation 4.6 bpm, against 5.4 (spline) and 5.5 (beats).

Data: beat times of the recording of slide 4, $K = 0.8\,\mathrm{rad/s}$, 300 s.

Artefact cleaning

Problem: one beat missed by the detector, one false extra beat. What do they do, and how do we find them?



Effect: a missed beat doubles one interval (IHR 35 bpm); an extra beat splits one (160 and 107 bpm). One such value dominates a mean, a standard deviation or a spectrum.

Find (nonparametric, L5): compare each interval with the median of the 11 intervals centred on it; flag it if the difference exceeds 5 MAD of these differences. Here MAD $= 0.049$ s: 3 intervals flagged.

Correct: a doubled interval $\rightarrow$ insert a beat in the middle; two short ones $\rightarrow$ remove the beat between them.

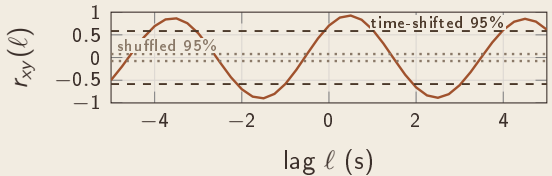
Report: 1 beat inserted, 1 removed, of 331.

Data: as on slide 5; one beat deleted (138 s) and one false beat inserted (201 s) by hand.

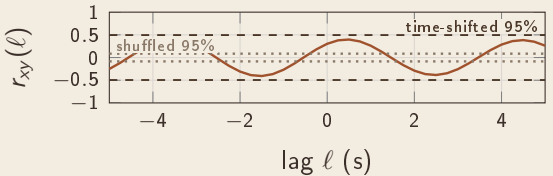
Cross-correlation, tested with surrogates

Problem: do the IHR and breathing change together, with what delay, beyond chance?

IHR and *its own* breathing



IHR and *another* recording's breathing



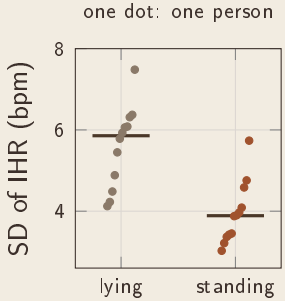
1. **Cross-correlation** $r_{xy}(\ell)$: correlation (L7) of $x(t)$ (IHR) with $y(t-\ell)$ (breathing) at each lag. Left: peak 0.93 at $\ell = 0.5$ s – the IHR follows breathing, 0.5 s later.
2. **Chance alone**: an unrelated pair (right: another recording's breathing) still reaches 0.41.
3. **Thresholds from surrogates** (H_0: no relation; threshold: 95th percentile, over surrogates, of the largest $|r_{xy}|$). *Dotted*, shuffled IHR: its temporal evolution destroyed, threshold ≈ 0.08 – even the unrelated pair looks "significant". *Dashed*, time-shifted breathing: both evolutions kept, timing random – left above, right below.

Surrogates must keep each signal's temporal evolution and randomise only what is tested.

Data: IHR (slide 6) and breathing, 280 s; right: breathing of a second recording.

Effect sizes with bootstrap intervals

Problem: does the IHR vary less when standing? Its standard deviation in 12 people lying and 12 *other* people standing: how large is the difference, and how sure are we?



Bars: group medians, 5.86 and 3.89 bpm.

Effect size: how large the difference is – not only whether there is one.

what it measures	value	95% interval
typical difference, standing − lying: median of all 12 × 12 differences (Hodges–Lehmann: the size estimate of the rank-sum test, L5)	−1.74 bpm	−2.45 to −0.88
chance that a random standing person varies less than a random lying one: $U/144$ of the rank-sum test (0.5: no difference)	0.92	0.80 to 1.00

95% interval by the bootstrap (L5): (1) draw 12 people at random from each group, with repeats; (2) recompute both effect sizes; (3) repeat 20 000 times – the middle 95% of the results is the interval. Resample people, who are independent – not values within one recording.

Data: 24 simulated recordings of 300 s, one per "person": 12 lying (K 0.6–1.2 rad/s), 12 standing, weaker coupling ($K\times$0.5–0.9).

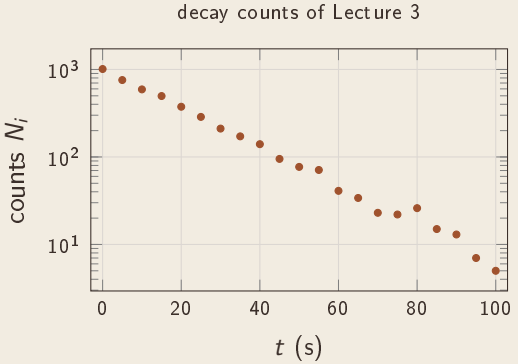
PART B

Bayes' theorem

What do data tell us about a model's parameters, and how sure can we be? Likelihood, prior, posterior, for one parameter.

Two questions about the same data

Problem: Part A showed that heart and breathing change together, but not how strongly one acts on the other: that strength is a **parameter of a model**. How do data tell us about a parameter, and how sure can we be? Simplest case first: the decay time τ of Lecture 3, where physics gives the model.



Data: N_i, decays counted in the 5-s interval starting at t_i. Model: expected count $\mu_i = N_0 e^{-t_i/\tau}$.

Lecture 3: which τ makes the data most probable?
One number: the best fit.

Today: how probable is each τ, given the data?
A distribution: what we know about τ.

For now $N_0 = 1000$ is taken as known.

Data: decay counts simulated in Lecture 3: $N_0 = 1000$, $\tau = 20$ s, Poisson noise.

Bayes' theorem

Problem: how do we turn "probability of the data given θ" into "probability of θ given the data"?

Example: a diagnostic test

10 000 people, 1 in 100 ill: 100 ill, 9 900 healthy.

The test is positive for 99% of the ill (99) and, wrongly, for 1% of the healthy (99).

Ill among the 198 positive: $99/198 = 50\%$.

$p(\text{pos.} \mid \text{ill}) = 99\%$, but $p(\text{ill} \mid \text{pos.}) = 50\%$: the **prior**, 1 in 100, matters.

Here θ: ill or healthy; d: a positive test.

$$p(\theta \mid d) = \frac{p(d \mid \theta)\, p(\theta)}{p(d)}$$

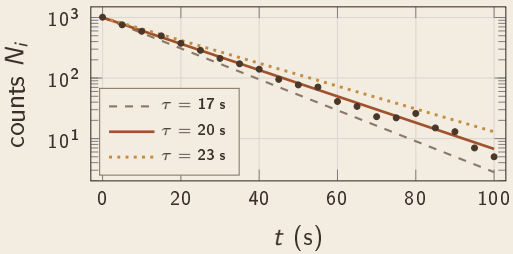
$p(\theta \mid d)$	**posterior**	what we know about θ after the data d
$p(d \mid \theta)$	**likelihood**	how probable the data are for each θ
$p(\theta)$	**prior**	what we know before the data
$p(d)$	**evidence**	makes the posterior integrate to 1

posterior $\propto$ likelihood $\times$ prior

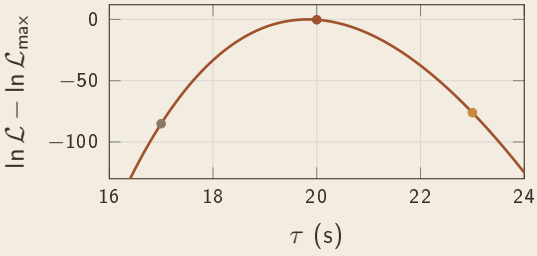
The likelihood: the model of the counts

Problem: how probable are the observed counts for a given τ?

each τ predicts a curve



how well each curve fits the counts



Expected count $\mu_i = N_0 e^{-t_i/\tau}$; counts of independent decays are Poisson (L5): $p(N_i \mid \tau) = \mu_i^{N_i} e^{-\mu_i} / N_i!$. Independent counts multiply: $\mathcal{L}(\tau) = \prod_i p(N_i \mid \tau)$, so $\ln \mathcal{L} = \sum_i (N_i \ln \mu_i - \mu_i) + \text{const}$. Its maximum, $\hat{\tau} = 19.83$ s, is the **maximum-likelihood** value (L3). At $\tau = 17$ or 23 s, $\ln \mathcal{L}$ is about 80 lower: those curves make the counts e^{80} times less probable.

Data: decay counts of Lecture 3.

The prior: what we knew before

Problem: the likelihood is not a distribution of τ. What else does Bayes' theorem need?



The **prior** $p(\tau)$ states, as a distribution, what we know before these counts.

- **flat:** only $\tau > 0$; the upper bound 100 s, far beyond any plausible value, only makes the prior normalisable
- **informative:** an earlier measurement, $\tau = 24 \pm 1.5$ s, Gaussian

The prior is part of the result

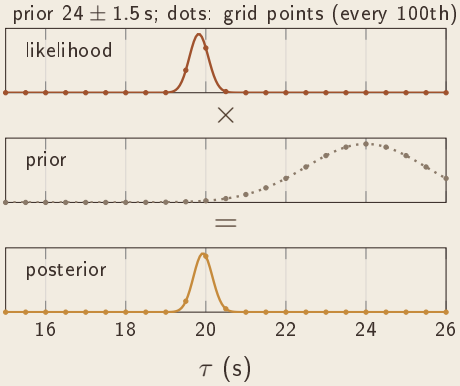
State it, and check how much the result depends on it. Prior knowledge enters twice, openly: in the model (likelihood) and in the prior.

The posterior on a grid

Problem: how do we compute the posterior in practice?

A grid: evenly spaced trial values of τ – here 2201 from 15 to 26 s, one every 0.005 s (outside, the posterior is negligible). Joined up, they give the posterior curve:

1. at each value: $\ln \mathcal{L} + \ln p(\tau)$
2. subtract the maximum, exponentiate
3. divide by the area: $\int p(\tau \mid d)\, d\tau = 1$



Flat prior: the posterior is the likelihood, normalised:

median	19.83 s
68% interval	19.59–20.06 s
95% interval	19.37–20.29 s

Data: decay counts of Lecture 3.

Credible and confidence intervals

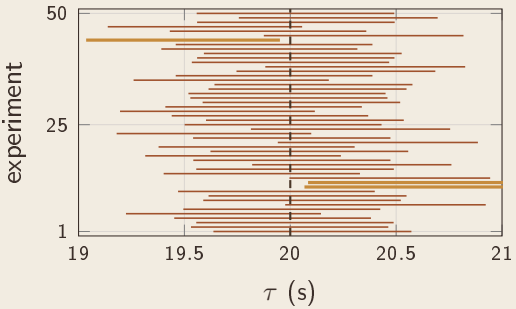
Problem: what does "95%" mean for the interval 19.37–20.29 s?

Credible interval (Bayesian)

Given these data and this prior, τ lies in it with probability 95%: a statement about τ, for *this* data set.

Confidence interval (frequentist)

Over many repeated experiments, the procedure gives intervals containing the true τ in 95% of them: a statement about the procedure.

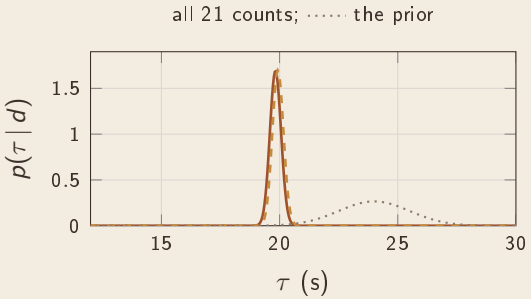


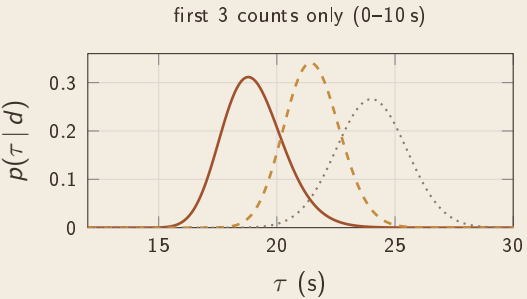
Check by simulation: 2000 experiments with $\tau = 20$ s (dashed); the flat-prior 95% credible interval contained 20 s in 94.5%: here it also works as a 95% confidence interval. Shown: the first 50, misses in ochre.

Data: 2000 new sets of counts simulated as in Lecture 3 ($\tau = 20$ s).

How much does the prior matter?

Problem: would a different prior change the conclusion?





median, 95% interval	flat prior	prior 24 ± 1.5 s
all 21 counts	19.83, 19.37–20.29	19.93, 19.47–20.39
first 3 counts	18.95, 16.69–21.83	21.48, 19.29–23.87

Many data: the data decide. Few data: the prior pulls the posterior; the true value ends near its edge.

Data: decay counts of Lecture 3; right, only the first three.

PART C

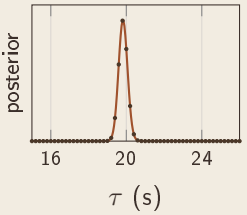
Sampling the posterior

With many parameters there are too many combinations to compute the posterior at each. Instead, a random walk visits each value as often as the posterior says: Markov chain Monte Carlo.

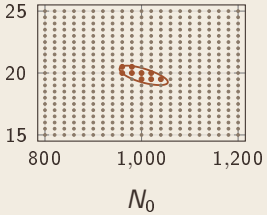
Why a grid fails in many dimensions

Problem: slide 14 used a grid of τ values. With N_0 unknown too, the grid covers a plane of pairs (N_0, τ). With m parameters, how many evaluations?

1 parameter:
a row of points



2 parameters: 9 of 441
inside the 95% line



m	grid points	at 10^6 per second
1	10^2	instant
2	10^4	instant
5	10^{10}	3 hours
10	10^{20}	3 million years

With e.g. 100 values per parameter: cost 100^m (L1), and most points lie where the posterior is almost zero.

Idea

Draw **samples** from the posterior instead: values of $\boldsymbol{\theta} = (N_0, \tau)$ that appear as often as their probability. Every summary is then a statistic of the samples, as in Monte Carlo (L2, L5).

Data: decay counts of Lecture 3; right: a 21×21 grid over N_0 800–1200, τ 15–25 s, flat priors.

The Metropolis–Hastings algorithm

Problem: how do we draw samples from a posterior we can evaluate only up to a constant?

Start at $\boldsymbol{\theta}_0$. At each step:

1. **propose** $\boldsymbol{\theta}' = \boldsymbol{\theta}_n +$ a random step (Gaussian, width $\Delta\boldsymbol{\theta}$, symmetric)
2. **accept** with probability

$$\min\left(1,\ \frac{p(\boldsymbol{\theta}' \mid d)}{p(\boldsymbol{\theta}_n \mid d)}\right)$$

3. if accepted, $\boldsymbol{\theta}_{n+1} = \boldsymbol{\theta}'$; if not, $\boldsymbol{\theta}_{n+1} = \boldsymbol{\theta}_n$ (counted again)

Why it works

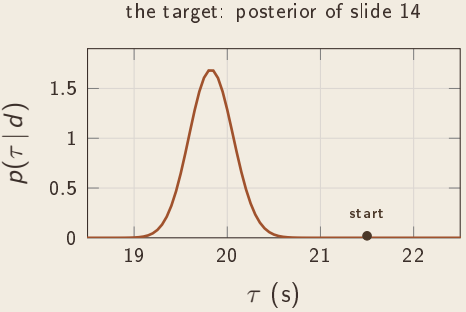
Twice as probable: always accepted; half as probable: half the time. This gives **balance**, $p(\boldsymbol{\theta})P(\boldsymbol{\theta} \to \boldsymbol{\theta}') = p(\boldsymbol{\theta}')P(\boldsymbol{\theta}' \to \boldsymbol{\theta})$: a chain that has the posterior distribution keeps it.

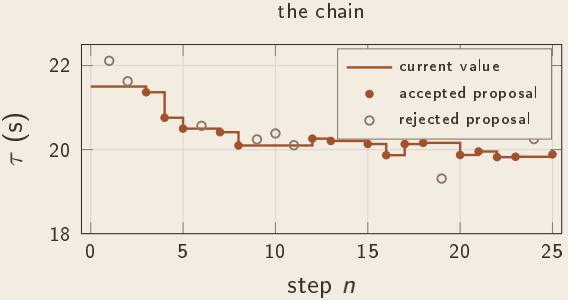
Only the **ratio** is needed: the evidence $p(d)$ cancels.

The samples form a **Markov chain**: each depends only on the one before. Successive samples are therefore not independent.

Metropolis–Hastings step by step

Problem: what does the chain actually do? The first 25 steps for τ alone ($N_0 = 1000$), steps of width 0.3 s.



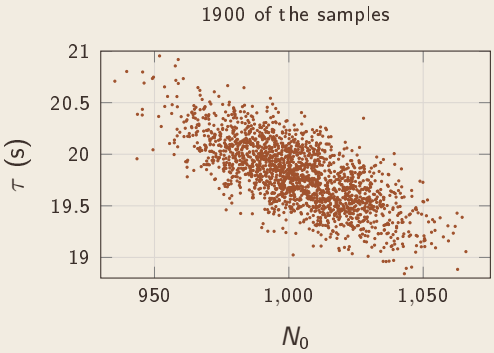


Start at 21.5 s, in the tail. Uphill proposals (towards 19.8 s) are always accepted, downhill ones sometimes. A rejected proposal (open circle) leaves the chain where it is: that value is counted again. 16 of 25 accepted; after about 20 steps the chain moves around the peak.

Metropolis–Hastings in action: N_0 and τ

Problem: both N_0 and τ unknown (flat priors, both positive), as in Lecture 3.





Start far away, $(N_0, \tau) = (500, 60\,\text{s})$; steps of width 15 and 0.4 s; 20 000 steps, 46% accepted. The transient from the start lasts about 350 steps: the first 1000 (**burn-in**, before the dashed line) are discarded. Right: the samples form the tilted cloud of a correlated posterior (L3).

Data: decay counts of Lecture 3.

Checks: step width and burn-in

Problem: the chain is a dynamical process we build, whose stationary distribution is the posterior. How do we choose its step width, and when is the transient from its start over?



Stationarity check: after the transient, any stretch of the trace has the same statistics; a slow drift means the transient is not over.

Step width sets the chain's time scale. Tiny steps: almost all accepted, each moves little – long correlation time (L6). Huge steps: almost all rejected – long stays at one value. Rule of thumb: about 44% accepted for one parameter, 23% for many, decorrelates fastest (Gelman, Roberts and Gilks, 1996). Here 48% in 1000 steps; 46% over 20 000 on slide 21.

Burn-in is the transient: from 60 s, about 350 steps to forget the start (slide 21). Discard it; here 1000 steps.

Data: decay counts of Lecture 3.

Results from the samples

Problem: what do we report, and is the sampler right?

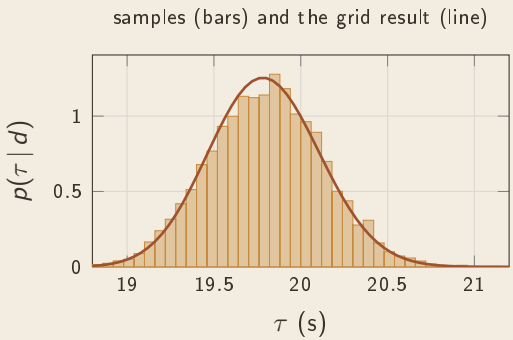
From the 19 000 samples:

	median	95% interval
τ (s)	19.81	19.17–20.44
N_0	1002	963–1045

Correlation of N_0 and τ: -0.70.

Standard deviation of τ: 0.33 s – the ± 0.34 s spread of the weighted fit in Lecture 3.

Check, grid of 221×241 pairs: τ 19.79, 19.18–20.42 s.



Two parameters: the grid is still affordable and confirms the sampler. Ten: only the sampler remains – test it first where the answer is known.

PART D

Outlook

When the parameters themselves change in time.

Dynamical Bayesian inference

Problem: two oscillators – like breathing and heartbeat – whose coupling changes in time.

Can we follow the coupling from the two phases?

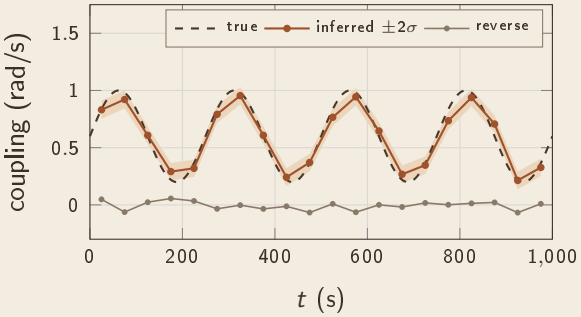
Model: for each oscillator, here the heart:

$$\dot{\varphi}_h = \sum_{k=1}^{7} c_k \, \Phi_k(\varphi_h, \varphi_r) + \varepsilon_h(t)$$

$\Phi_1 = 1$ (c_1: the frequency ω_h); sine and cosine of φ_h, φ_r, $\varphi_h - \varphi_r$. The coefficient of $\cos\varphi_r$ is the coupling $K(t)$; ε_h: Gaussian noise.

Inference in 50-s windows: linear in the c_k, Gaussian noise, so the posterior is Gaussian and exact, as for least squares (L3). Each window's posterior, widened, is the next prior.

simulated: respiration → heart



Inferred coupling follows $K(t)$ (18 of 20 windows within $\pm 2\sigma$); reverse: zero within $\pm 2\sigma$ (20 of 20).

Data: the model of slide 4, coupling $K(t)\cos\varphi_r$ with $K(t) = 0.6 + 0.4\sin(2\pi t/250\,\mathrm{s})$, 1000 s.

What dynamical Bayesian inference adds

Problem: what does this give that a single fit does not, and what does it assume?

It gives

- parameters **followed in time**: a nonautonomous system (L2), not noise
- the **direction** of coupling, within the assumed model
- other parameters (here frequencies) and noise strength, with uncertainties

It assumes

- a model: equations for phases (as here), amplitudes or full states, with chosen Φ_k
- Gaussian white noise
- parameters nearly constant within one window

Correlation is not coupling: r_{xy} (slide 7) has no direction, its size depends on the noise, and it averages over time.

Whole record at once: 0.59 rad/s – an average that hides a fivefold change.

Summary

Step	Tools	Check
Beat times to a series	RR intervals (HRV), IHR, interpolation to an even grid	the interpolant (cubic, not linear)
Artefacts	running median; flag beyond 5 MAD	what is flagged; report corrections
Relation of two signals	cross-correlation; largest $\lvert r_{xy} \rvert$	surrogates that keep each signal's temporal evolution (time shift)
Size of a change	effect sizes with bootstrap percentile intervals	resample independent units
Model the data	likelihood (Poisson, Gaussian $\leftrightarrow \chi^2$)	does physics supply it?
State prior knowledge	flat or informative prior	repeat with another prior
Posterior, few parameters	posterior at evenly spaced values (grid); median and credible interval	coverage by simulation
Posterior, many parameters	Metropolis–Hastings	transient (burn-in); stationarity; a known case
Parameters in time	dynamical Bayesian inference	window length; model

Before you report: six questions

1. Did my preparation change the data – interpolant, artefact corrections – and did I report it? *(Part A)*
2. Do my surrogates keep everything except the property tested, and do my resamples draw independent units? *(time shift; people)*
3. Does physics supply the model of the data? *(otherwise: nonparametric methods, L5)*
4. What is my prior, and does the result change with another one? *(few data: the prior decides)*
5. Is my chain's transient over, and is it stationary? *(burn-in; stationarity)*
6. Does the sampler reproduce a case with a known answer? *(verification)*

Report the posterior: median, interval, correlations – and the prior.

TO TAKE AWAY

Posterior $\propto$ likelihood $\times$ prior.

Before any model: prepare the data openly, test relations with surrogates that keep each signal's temporal evolution, and report the size of an effect with its interval.

With a model that physics supplies: the result is a distribution, not a number – sample it when a grid is too costly, and check the sampler where the answer is known.